

Challenge (Habanero)

- Q1.** A hiker's footsteps echo 2^x times per minute. If the echoes are heard 16 times, what is x ?
- (A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q2.** The fuel consumption of a vehicle is $3 \cdot 2^{t-2}$ liters after t hours. What is the fuel consumption after 5 hours?
- (A) 48 liters
(B) 36 liters
(C) 24 liters
(D) 12 liters
(E) 6 liters
- Q3.** A species of rabbit has a population that grows by a factor of $2^{1/3}$ every month. If there are currently 64 rabbits, how many were there 6 months ago?
- (A) 8
(B) 16
(C) 32
(D) 4
(E) 2
- Q4.** A waterfall flow rate is given by the equation $4 \cdot 3^{x/4}$ liters per second. What is the flow rate when $x = 16$?
- (A) 144 liters
(B) 108 liters
(C) 72 liters
(D) 48 liters
(E) 36 liters
- Q5.** A mountain guide's income is $500 \cdot 2^{n-3}$ dollars per year, where n is the number of years they have been guiding. What is their income after 7 years?
- (A) 8000
(B) 4000
(C) 2000
(D) 1000
(E) 16000
- Q6.** If $2^{x-4} = 32$, then what is x ?
- (A) 6
(B) 7
(C) 5
(D) 3
(E) 4
- Q7.** A forest's tree density is modeled by $5 \cdot 2^{d/5}$ trees per acre, where d is the depth of the forest in miles. What is the tree density at a depth of 10 miles?
- (A) 160 trees
(B) 80 trees
(C) 40 trees
(D) 20 trees
(E) 10 trees
- Q8.** A hiker's elevation gain is $3 \cdot 2^{e-1}$ feet, where e is the number of hours they've been hiking. What is their elevation gain after 6 hours?
- (A) 96 feet
(B) 48 feet
(C) 24 feet
(D) 192 feet
(E) 384 feet

Open Ended Questions (FRQ)

- Q9.** A park ranger is monitoring a herd of deer. The population grows by a factor of $2^{1/2}$ every 2 months. If there were initially 20 deer, how many deer will there be in 1 year? Show your work and justify your answer.
- Q10.** A natural spring's water level is given by the equation $10 \cdot 2^{w/10}$ feet, where w is the amount of rainfall in inches. If it has been raining for 5 hours and the rainfall rate is 2 inches per hour, what is the current water level? Show your work and justify your answer.

Chef's Tips

- Make sure to review the rules of exponentiation before starting the test
- Use the given formulas to solve the problems
- Check your work carefully to avoid errors